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Artificial Intelligence Qualifying Exam

1. Show the comparison — ~~cost~~ uniform cost search, Greedy best first search and A* Search.

Ans. These are informed/uninformed search algorithms used in Artificial Intelligence to find a path from an initial state to a goal state.

A. ~~Cost~~ Uniform Cost Search:

Uniform cost search expands the node with the lowest total path cost from the starting node. It uses the cost function: $f(n) = g(n)$

$g(n)$: cost from the starting node to the current node

It guarantees an optimal solution if all steps ~~are~~ costs are positive.

It does not use a heuristic.

B. Greedy Best-first Search:

Greedy best first search expands the node that appears closest to the goal based on a heuristic function.

$$f(n) = h(n)$$

$h(n)$: Estimated cost from the current node to the goal.

It is generally faster but does not guarantee the optimal solution.

It depends on the quality of the heuristic.

c. A* Search:

A* search combines the actual path cost and estimated remaining cost to find the least path.

$$f(n) = g(n) + h(n)$$

$g(n)$: Actual cost from the starting node

$h(n)$: estimated cost to reach the goal.

It is complete and optimal when the heuristic admissible and the standard search conditions are satisfied.

2. Evaluation Metrics for these search algorithms and how they are set:

(i) Completeness: Whether the algorithm can find a solution if one exists.

(ii) Optimality: whether the algorithm finds the lowest-cost solution.

(iii) Time complexity — How much time or how many nodes are required to find the solution.

(iv) Space complexity: How much memory is required to store generated or explored nodes.

How they are set —

⇒ Completeness and optimality are determined by the algorithm and it's assumptions.

⇒ Time complexity is measured by counting expanded or generated nodes.

⇒ Space complexity is measured by counting the maximum number of nodes stored in memory.

⇒ The heuristic $h(n)$ and path cost $g(n)$ determine which nodes are selected for expansion.

3. Bookstore Dataset : Entropy calculation

Suppose a bookstore wants to predict whether a visitor will purchase a book based on their device, age group and browsing time.

Dataset

Visitor ID	Device	Age Group	Browsing Time	Purchased
1	Mobile	Young	Short	No
2	Mobile	Young	Long	Yes
3	Desktop	Adult	Short	No
4	Mobile	Adult	Long	Yes
5	Desktop	Young	Long	Yes
6	Desktop	Adult	Short	No
7	Mobile	Senior	Long	Yes
8	Desktop	Senior	Short	No

There are 8 visitors:

Yes (Purchased) = 4

No (Purchased) = 4

A. Calculate Entropy:

Entropy measures the impurity or uncertainity in a dataset.

formula:

$$H(S) = -\sum P_i \log_2(P_i)$$

where:

P_i = Probability of each class

Yes = $4/8 = 0.5$

No = $4/8 = 0.5$

$$H(S) = -[(0.5)\log_2(0.5) + (0.5)\log_2(0.5)]$$

$$= -[0.5(-1) + 0.5(-1)]$$

$$= 1$$

The initial entropy is 1 bit.

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4. Calculate Information Grain:

Information Grain measures how much uncertainty is reduced after splitting the dataset using an attribute.

Formula:

$$IG(S, A) = H(S) - H(S|A)$$

where:

$H(S)$ = Original entropy

$H(S|A)$ = Weighted entropy after splitting by ~~at~~ attribute A.

A. Information gain for device:

Mobile: 4 visitors (3 Yes, 1 No)

$$H(\text{Mobile}) = -\left[\frac{3}{4}\log_2\frac{3}{4} + \frac{1}{4}\log_2\frac{1}{4}\right]$$

$$= 0.8113$$

Desktop: 4 visitors (1 Yes, 3 No)

$$H(\text{Desktop}) = -\left[\frac{1}{4}\log_2\frac{1}{4} + \frac{3}{4}\log_2\frac{3}{4}\right]$$

$$= 0.8113$$

Weighted entropy:

$$H(S|\text{Device}) = \frac{4}{8}(0.8113) + \frac{4}{8}(0.8113)$$

$$= 0.8113.$$

Information Grain: $1 - 0.8113 = 0.1887$

$\therefore IG(\text{Device}) = 0.1887.$

B. Information Gain for Age Group:

Young: 3 visitor (2 Yes, 1 No)

$$H(\text{Young}) = -\left[\frac{2}{3}\log_2\frac{2}{3} + \frac{1}{3}\log_2\frac{1}{3}\right]$$

$$= 0.9183$$

Adult: 3 visitor (1 Yes, 2 No)

$$H(\text{adult}) = 0.9183$$ [As we saw in previous calculation]

Senior: 2 visitors (1 Yes, 1 No)

$$H(\text{Senior}) = 1$$

Weighted Entropy:

$$H(S|\text{Age}) = \frac{3}{8}(0.9183) + \frac{3}{8}(0.9183) + \frac{2}{8}(1)$$

$$= 0.9387$$

Information Gain:

$$IG(\text{Age}) = 1 - 0.9387$$

$$= 0.0613$$

C. Information Gain for Browsing Time:

Short: 4 visitors (0 Yes, 4 NO)

$H(\text{Short}) = 0$

Long: 4 visitors (4 Yes, 0 NO)

$H(\text{Long}) = 0$

weighted entropy:

$H(\text{Time}) = \frac{4}{8}(0) + \frac{4}{8}(0) = 0$

Information gain:

$IG(\text{Time}) = 1 - 0 = 1$

5. Which attribute should be the Root node? Justify your answer.

Ans: The root node should be browsing time, because it has the highest information gain. As we have seen in the previous section, ~~for browsing time~~ information gain for browsing time is 1, which is higher than the information gain for both device and age group.

Justification: Browsing time has the highest information Gain (1). It divides the dataset into two pure groups:

Short browsing time → No purchase
Long browsing time → Yes purchase

Therefore, it reduces entropy from 1 to 0 and is the best attribute for the root node.

6. Cooling Rate, Initial Temparature and Stopping condition.

Ans: These are important parameters for Simulated Annealing, an optimization algorithm inspired by the cooling process of metals.

A. Initial Temparature:

Initial temparature (T_0) is the starting temparature of the algorithm. A high initial temparature allows the algorithm to accept more worse solutions to explore different possibilities. Ex. $T_0 = 1000°c$

B. Cooling Rate:

Cooling rate determines how quickly the temperature decreases during the search process.

formula: $T_{new} = \alpha \times T_0$

Where: T_0 = Previous Temperature / Initial Temperature

α = Cooling Rate (usually between 0 and 1)

T_{new} = New Temperature

Ex: If $T_0 = 1000°c$ and $\alpha = 0.9$

Then $T_{new} = 1000°c \times 0.9 = 900°$

The new Temperature 900°c

C. Stopping Condition:

The stopping condition determines when the algorithm should terminate.

Common stopping Conditions are:

1. Temperature reaches a minimum threshold.
2. Maximum number of iteration is completed.
3. No significant improvement occurs after several iterations.

Ex: Stop when 1000 iterations are completed.